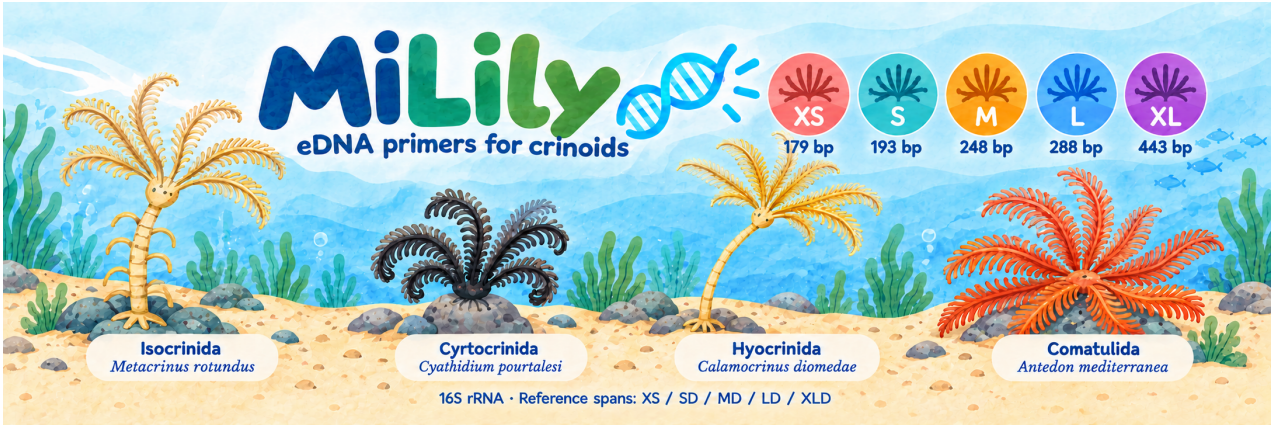
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MiLily, a set of universal PCR primers for metabarcoding environmental DNA from sea lily

Fei Xia (GUIBA-EX)

Department of Integrated Bioscience, Graduate School of Frontier Sciences, The University of Tokyo, Kashiwa 277-8561, Chiba, Japan

GitHub repository | 1738314330@edu.k.u-tokyo.ac.jp



Abstract

MiLily comprises candidate mitochondrial 16S rRNA primers for metabarcoding crinoid environmental DNA across five windows: XS, S, M, L and XL. Target matching was evaluated on a common panel of 27 references representing 25 taxonomic labels, with cross-matching screened against 7,177 non-target records. The low-degeneracy XS pair produced predicted amplicons of 159-194 bp, with 100% coverage and 96% exact matching; MD achieved 100% coverage and 84% exact matching. XS also matched 18 non-target records under the strict target criterion, including 10 exact sea-urchin matches. S-XL offer ordinary, complementary A/B and degenerate D alternatives at shared coordinates. XS prioritises short products and low degeneracy, whereas MD combines complete panel coverage with fewer non-target matches, providing alternatives for different sample conditions.

Background and purpose

MiLily targets mitochondrial 16S rRNA in Crinoidea, including stalked sea lilies and feather stars. Molecular and phylogenomic studies reveal complex relationships between body plans and ancestry, supporting lineage-aware reference sampling. Rouse et al., 2013; Rouse et al., 2026

The design follows the MiFish principle of conserved primer-binding regions flanking an informative short marker. Five length windows allow comparison of target matching, marker length and background hits. Miya et al., 2015

1. Methods

Reference data and matching criteria

Coordinates are 1-based on the extracted, ungapped AF049132.1 16S sequence. This reference represents *Florometra serratissima*, whose complete mitochondrial genome and gene arrangement were described by Scouras and Smith. Scouras & Smith, 2001 The target FASTA contains 269 records. Headline comparisons use a frozen common panel of 27 references and 25 taxonomic labels, weighted equally by label and then within label, to control for differences in partial-reference availability among windows.

Coverage requires ≤2 mismatches per primer and IUPAC-compatible terminal 3 nt; exact matching requires zero mismatches at both ends. The background contains 7,177 non-target records. WC3, GT2 and WC4 count paired hits with ≤3 mismatches per primer, ≤2 under a G-T pairing model, and ≤4, respectively; these permissive screens do not require terminal-3 compatibility. Strict background matching separately applies the target coverage criterion. Non-target hits are reported as reference-record counts.

Candidate design and verification

Ordinary, A, B and D alternatives share identical forward and reverse boundaries within each S-XL window. Single pairs are preferred where sufficient. A/B denote separate-PCR unions, retained only when both components contribute unique coverage or exact matches. D denotes a single degenerate pair. XS was evaluated independently as a supplied fixed pair.

Terminal checks require no terminal homotriplet, at least one GC in the final 3 nt, 1-4 GC in the final 5 nt and no homopolymer longer than four bases; 3′ degeneracy is allowed. Duplex, hairpin and dimer thermodynamics are calculated with Primer3. Untergasser et al., 2012 Calculations use 50 mM monovalent salt, 1.5 mM Mg^{2+} and 0.6 mM dNTP, at 50 nM per variant and 200 nM total per direction. S-XL require structure Tm <55°C and a forward/reverse duplex-Tm midpoint difference ≤5°C. WC3 and GT2 limits are 20 records each for SD/MD and five each for other S-XL portfolios. XS is evaluated separately as a fixed-sequence scheme.

Exhaustive full-sequence scans check both strands, paired orientation and distance, followed by independent algorithmic verification. Full parameters and execution instructions are available in Methods and Reproduction.

Marker selection and reference curation

Short-marker selection considers both DNA recovery and taxonomic information. Mini-barcode and window-selection studies provide the basis for evaluating these tradeoffs, while ecoPrimers formalises coverage and discrimination as complementary criteria. Meusnier et al., 2008; Boyer et al., 2012; Riaz et al., 2011

Crinoid reference curation draws on molecular classification and integrative identification of Comatulidae. Summers et al., 2014; Summers et al., 2017

2. Primer sequences

Table 1. All 17 primer pairs. Sequences are written 5′ to 3′. Ordinary, A, B and D alternatives share coordinates within each S-XL window. A/B indicates separate-PCR unions; D is a degenerate single pair. XS is the fixed short pair. F/R values give numbers of concrete sequence variants.

Name	Primer sequences (5′ to 3′)	F/R variants
XS	F TTGGGGCAACCDCGAAG R CAACATCGAGGTCGCAAAC	3/1
SD	F GATCTAMTGAWRYYCCAA R TRTCCCTTTTRGYYTWGY	32/64
S	F GATCTAATGAAACCCCAA R TATCCCTTTTGGTTTAGC	1/1
SA	F GATCTAATGAAACCCCAA R TATCCCTTTTGGTTTAGC	1/1
SB	F GATCTAATGAAACCCCAA R TGTCCCTTTTGGTTTAGC	1/1
MD	F TTACTGTTRCAAAAGTAAAYWAAGAR R GRCTGCTTYTRGGCCTAC	16/8
M	F TTACTGTTGCAAAAGTAAACTAAGAG R GGCTGTTTCTAGGCCTAC	1/1
MA	F TTACTGTTACAAAAGTAAACTAAGAG R GGCTGCTTCTAGGCCTAC	1/1
MB	F TTACTGTTGCAAAAGTAAACTAAGAG R GACTGCTTCTAGGCCTAC	1/1
LD	F CAAGAGATCTAATGAAACCCCA R GRCTGCTTYTRGGCCTAC	1/8
L	F CAAGAGATCTAATGAAACCCCA R GGCTGCTTCTAGGCCTAC	1/1
LA	F CAAGAGATCTAATGAAACCCCA R GGCTGCTTCTAGGCCTAC	1/1
LB	F CAAGGGATCTAATGAAACCCCA R GGCTGCTTCTAGGCCTAC	1/1
XLD	F AGGCAAVACGARRRCTYAACTG R GACTYDCGTCGGTCTGAA	48/6
XL	F AGGCAAGACGAAAGCTTAACTG R GACTTGCGTCGGTCTGAA	1/1
XLA	F AGGCAACACGAAAACTTAACTG R GACTTGCGTCGGTCTGAA	1/1
XLB	F AGGCAAGACGAAAGCTTAACTG R GACTTGCGTCGGTCTGAA	1/1

IUPAC codes used: M=A/C; R=A/G; W=A/T; Y=C/T; V=A/C/G; D=A/G/T.

3. Target matching

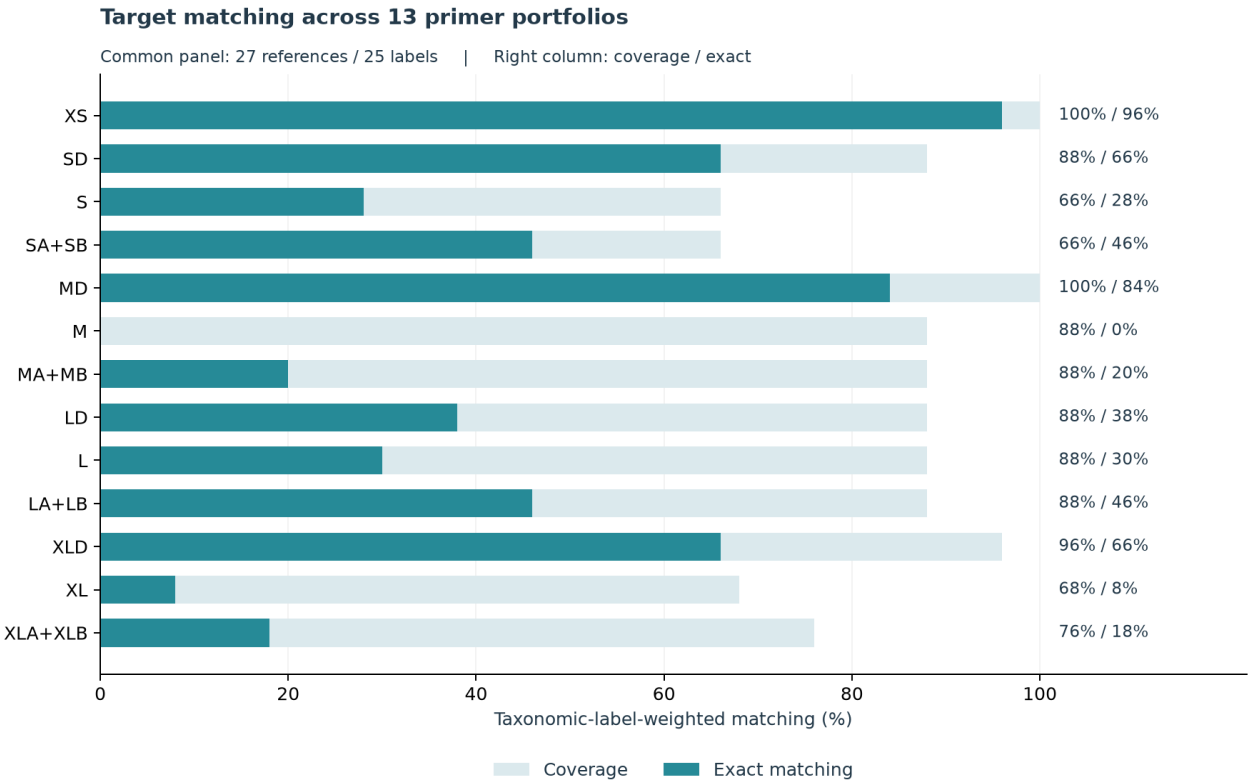


Figure 1. Label-weighted coverage and exact matching on the common 27-reference, 25-label panel. Pale bars show coverage, dark bars exact matching. Values at right are coverage / exact percentages. A/B values are unions of separate PCRs.

XS and MD cover the full common panel, with exact matching of 96% and 84%, respectively. Complementary S, M and L portfolios mainly increase exact matching; XLA+XLB also increases coverage relative to XL.

Independent verification covers 918 target decisions and 34 background-set checks with zero disagreements. A wider product-length scan leaves the XS matching metrics unchanged.

Base-by-base alignments for all pairs are available in the repository alignment report and SVG figure directory.

4. Background matching

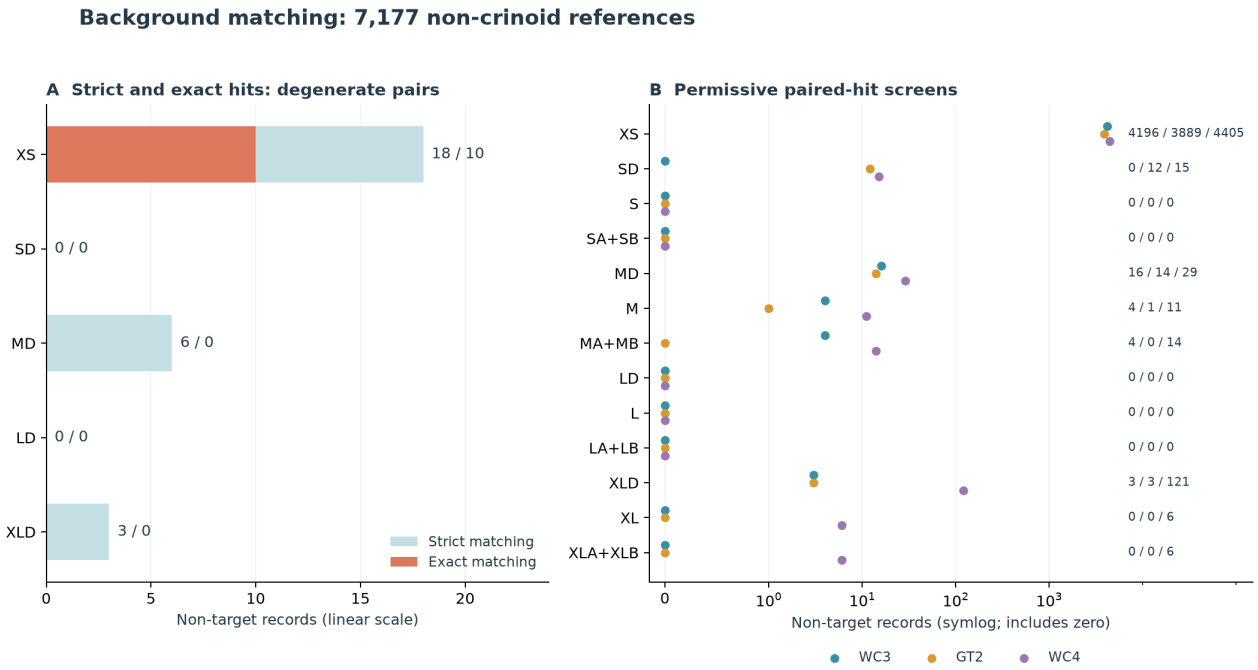


Figure 2. Non-target matching against 7,177 non-crinoid references. A: strict / exact record counts for the degenerate single pairs. Strict matching requires ≤2 mismatches per primer and compatible terminal 3 nt. B: WC3 / GT2 / WC4 permissive paired-hit counts for all portfolios, with a symlog axis that includes zero.

XS gives 18 strict non-target hits, including 10 sea-urchin references matching both primers exactly. Their product sizes overlap those of crinoids. SD, MD, LD and XLD give 0, 6, 0 and 3 strict hits, respectively. Background-screen counts describe matching reference records.

5. Product length and primer complexity

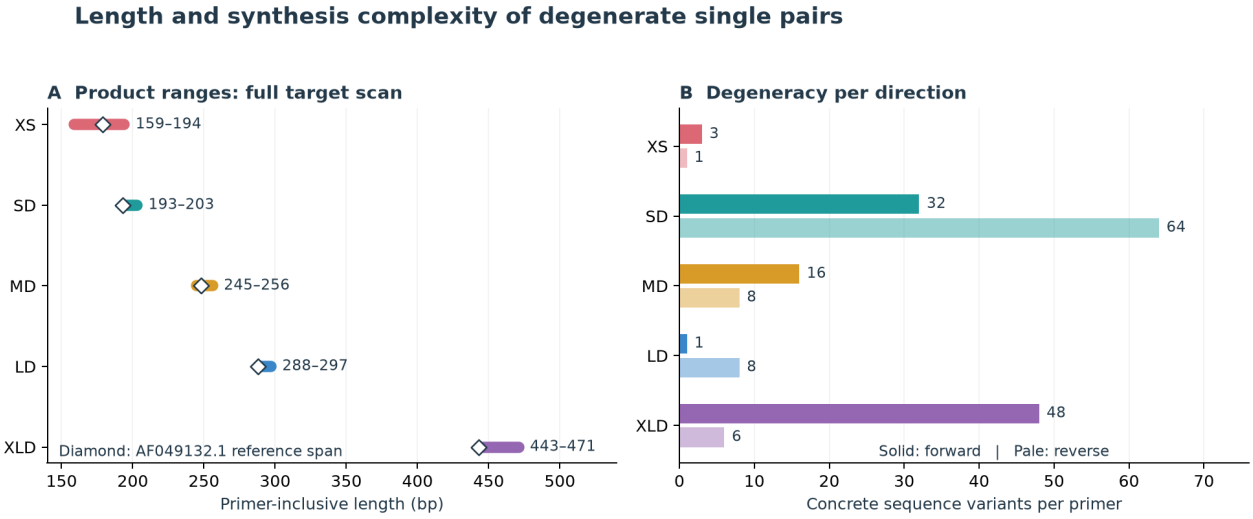


Figure 3. A: primer-inclusive product ranges from the full target-reference scan; diamonds indicate AF049132.1 reference spans. B: concrete variants per primer, with solid F and pale R bars. XS spans 159-194 bp in the full target scan and 178-194 bp in the common panel.

XS structure assessment

XS passes terminal rules and duplex-Tm compatibility. Its three F variants predict internal short hairpins with Tm of 62.6-69.4°C, above the 55°C S-XL design threshold. The primer 3′ end lies outside the stem; predicted hairpin ΔG at 57°C is approximately -0.35 to -0.65 kcal/mol. The fixed XS pair is evaluated alongside the constrained S-XL designs.

Practical comparison

XS provides the shortest product and low degeneracy (3/1). MD combines complete common-panel coverage with fewer permissive background matches. SD has higher degeneracy (32/64); LD has the simplest F sequence among the S-XL degenerate options (1/8). These comparisons support selecting a small set of assays for parallel experiments.

6. Discussion and collaboration

XS achieves 100% coverage and 96% exact matching with F/R degeneracy of 3/1, making it a priority for short-amplicon testing. MD also covers the complete common panel and yields fewer permissive background hits. The two schemes offer different tradeoffs: XS emphasises short products and low degeneracy, whereas MD balances coverage and background discrimination. Of the 18 strict non-target XS hits, 10 sea-urchin records match both primers exactly and have predicted products overlapping the crinoid size range. Mixed samples containing sea-urchin DNA are therefore a useful test of competitive amplification.

MiFish accommodates lineage-specific binding variation with an additional primer variant, a principle relevant to complementary MiLily combinations. Miya et al., 2015 In MiLily, S, M and L unions primarily improve exact matching, whereas the XL union also improves coverage. Retention depends on each component's unique contribution. Shared coordinates among ordinary, A, B and D alternatives keep within-window comparisons focused on binding-sequence differences.

XS has 22/25 labels without observed shared barcodes (88%) on the common panel, the same as MD. Discrimination depends on the variation captured by a window, supporting joint evaluation of marker length and barcode information. Boyer et al., 2012 The present results are computational evaluations on shared reference data; subsequent tissue, defined-mixture and environmental tests will compare amplification efficiency and detected composition. Controlling mixture inputs is particularly useful because metabarcoding experiments have found substantial read-count differences among taxa with similar biomass. Elbrecht & Leese, 2015

Expanding the crinoid reference library requires both lineage representation and taxonomic reconciliation. Integrative work on Comatulidae found that external morphological differences and COI divergence do not follow a uniform relationship. Summers et al., 2017 For MiLily, this supports prioritising references linked to voucher specimens, morphological identifications and traceable sequences, while using the phylogenomic framework to distribute validation samples across major lineages. Rouse et al., 2026

Collaboration

We welcome collaborators interested in crinoid diversity, eDNA monitoring, primer design and reference-library development, **especially researchers interested in wet-lab validation**.

Potential collaborations include tissue-DNA amplification, defined mixtures and environmental samples, annealing-condition optimisation, and comparisons of amplification efficiency, taxon detection and non-target co-amplification among primer schemes. Contributions of reference sequences linked to specimen and identification records, experimental feedback and analytical suggestions are also welcome.

Please contact us at 1738314330@edu.k.u-tokyo.ac.jp with a brief description of your research interests, available materials and proposed work so that we can discuss experimental design and contributions together.

Data, code and permissions

Repository: https://github.com/GUIBA-EX/MiLily. This method document is based on repository commit 5e45893. Primer tables, reference records, base-matching figures, performance data and reproduction scripts are available there. Copyright © 2026 Fei Xia (GUIBA-EX). Reuse permissions are described in the repository copyright statement.

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